

Microcanonical Variational Theory of Radical Recombination by Inversion of Interpolated Partition Function

Part 2.— $CX_3 + O_2$ ($X = H, F, Cl$)

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The title theory (MVIPF) has been applied to the recombinations of O_2 with the radicals CH_3 , CF_3 , CF_2Cl , $CFCl_2$ and CCl_3 and compared, with generally good results, to experimental data which are available for all radicals but one (CF_2Cl). In particular, MVIPF, which is based on minimum information and uses an adjustable parameter (c) in a Gaussian-type switching function, correctly reproduces the fact that the recombination $CH_3 + O_2$ has a positive temperature coefficient, while all the other radical recombinations have a negative temperature coefficient. Consolidated results by MVIPF for the nine recombination reactions for which experimental data are available (the title reactions, plus $CH_3 + H$, $CH_3 + CH_3$, $CH_2Cl + CH_2Cl$, $CHCl_2 + CHCl_2$, $CCl_3 + CCl_3$), yield a linear relation for a function of the logarithm of c , which in principle allows one to predict a recombination rate constant in a series of similar reactions within a factor of two.

Radical recombination reactions of the type $CX_3 + O_2 \rightarrow CX_3O_2$ ($X = H, Cl, F$ and combinations thereof), have been the subject of several experimental studies, the most recent of which^{1,2} were done at a number of temperatures and pressures. Apart from their importance in connection with atmospheric chemistry, these reactions are also of considerable interest from a theoretical point of view, in that the high-pressure recombination rate constant of the first member of the series ($X = H$) presents a positive temperature coefficient, while the recombination rate constants of the other members show a negative temperature coefficient of varying magnitude.

Radical recombinations, including the title reactions, which proceed without formal activation energy, have to be dealt with within a variational formalism which, in a recent elaborate implementation, requires the calculation of much detailed microscopic information: witness the flexible transition-state theory (FTST) approach of Wardlaw and Marcus.³ However even the considerable computational effort of FTST does not seem able to reproduce the temperature coefficient of the recombination⁴ $CH_3 + H \rightarrow CH_4$, although it does deal successfully with the negative temperature coefficient of the recombination⁵ $CH_3 + CH_3 \rightarrow C_2H_6$. Note that the statistical adiabatic channel model (SACM)⁶ returns in virtually every case a positive temperature coefficient for the high-pressure rate constant of a radical recombination, in particular for the first two recombinations of the present series ($X = H$ and F) for which SACM calculation is available.⁷

Thus the challenge represented by the title reactions is two-fold: available microscopic information about the reactions is limited to basic molecular parameters of reactants and products, and the temperature coefficient does not seem to fit any obvious pattern. It seemed therefore useful to apply to this particular type of recombination the recently formulated microcanonical variational theory by inversion of interpolated partition function (MVIPF)⁸ with the object being to see if the theory is able to rationalize the seemingly disparate results and offer a coherent picture of this class of reactions. Including the four recombinations discussed in this paper, there are now a total of nine radical recombinations that

have been treated by MVIPF, a number sufficiently large to warrant taking stock. Hence in addition to the title reactions, the discussion will deal with all the other five MVIPF recombinations as well.

MVIPF is based on the proposition that for the purpose of calculating thermal rate constants there is actually enough information for a reasonably successful variational calculation in just the properties of reactants and products, without the need for a fully fledged potential-energy surface. This makes feasible the systematic treatment of a series of reactions of complex reactants for which potential-energy surface calculations would be prohibitive in time and expense.

MVIPF obtains the microcanonical rate constant for dissociation [eqn. (1) below] by inversion of a partition function interpolated between that of reactant and products, by means of a switching function. Thus the theory requires only modest computational effort, based as it is entirely on (equilibrium) properties of reactant and products, plus an adjustable parameter in the switching function. While the MVIPF performance tested out fairly well in the recombination $CH_3 + H$, and very satisfactorily in the recombination $CH_3 + CH_3$ (see ref. 8), the title reactions offer another, and perhaps more severe, test. Given below is a brief outline of the theory; details may be found in ref. 8.

Theory

The basic element is $k(E, J)$, the microcanonical rate constant for the decomposition of a molecule with specified internal energy E and specified angular momentum quantum number J , which is calculated from the variational version of the RRKM expression

$$k(E, J) = \frac{G^*(E^*, J, r_{\min})}{hN(E)} \quad (1)$$

Here the numerator represents the minimized state count for the transition state, localized at value $r_{\min}$ of the reaction coordinate, at specified J and energy E^* in excess of its own ground state (* is used to denote properties of transition

state), while $N(E)$ in the denominator is the density of internal states of the reactant molecule at energy $E = E^* + E_f(r_{\min})$, where $E_f(r_{\min})$ is the minimum energy for decomposition at J and $r_{\min}$. The J - and pressure-dependent thermal unimolecular rate constant $k_{\text{uni}}(J)$ for decomposition then follows from the standard expression

$$k_{\text{uni}}(J) = \frac{1}{Q_M} \int_{E_f(r_{\min})}^{\infty} \frac{\omega k(E, J) N(E) \exp(-E/kT) dE}{\omega + k(E, J)} \quad (2)$$

where $k(E, J)$ is given by eqn. (1) above, Q_M is the molecular partition function given by $Q_M = \int_0^{\infty} N(E) \exp(-E/kT) dE$ and ω is an effective collision frequency.

In a thermal reaction, the distribution of J is

$$P(J) = (2J + 1) \exp(-E_{\text{rot}}/kT) \quad (3)$$

where

$$E_{\text{rot}} = J(J + 1)B_e \quad (4)$$

B_e is the rotational constant of the molecule which is defined below. The J -averaged thermal rate constant for dissociation is then

$$\langle k_{\text{diss}} \rangle = \left[\sum_J k_{\text{uni}}(J) P(J) \right] / \sum_J P(J) \quad (5)$$

Finally the J -averaged thermal rate constant for recombination (which is identified with the experimental value of the recombination rate coefficient) follows *via* the equilibrium constant K_{eq} (which requires knowledge of the enthalpy of reaction ΔH_0 at 0 K).

In MVIPIF, it is assumed at the outset that, in addition to ΔH_0 , the only information available is the molecular parameters of reactant and products, sufficient for calculating the equilibrium constant and the vibrational-rotational partition functions of the reactant molecule Q_m [which is Q_M of eqn. (2) modified to exclude the reaction coordinate], and the partition function of the products Q_p . The partition function of the transition state, as a function of distance r along the reaction coordinate [$Q^*(r)$], is obtained by interpolation of the logarithms of the appropriate reactant and product partition functions

$$\ln Q^*(r) = (\ln Q_m)S(r) + (\ln Q_p)[1 - S(r)] \quad (6)$$

where $S(r)$ is a switching function such that $S(r) = 1$ at r_e (value of r at equilibrium, *i.e.* in the reactant molecule), and $S(r) = 0$ at $r \rightarrow \infty$ (*i.e.* in the products). Inversion of $Q^*(r)$ yields the r -dependent vibrational-rotational state count $G^*(E^*, J, r)$. [The J -dependence of $G^*(E^*, J, r)$ arises *via* E^*]. Thus the reactant-like state count, which involves essentially vibrations, connects smoothly with the product-like state count which involves both vibrations and rotations. In this way the difficult problem of the progressive conversion of quantized vibrations into classical rotations (the so-called 'transitional' modes) is dealt with without explicit reference to dynamics. The inversion procedure is briefly described below; for details see ref. 8.

If the transform parameter $s = 1/kT$ is shown explicitly, the transition-state partition function [eqn. (6)] becomes $Q^*(s, r)$. The state count $G^*(E^*, J, r)$ is given by the complex inversion integral

$$G^*(E^*, J, r) = \frac{1}{2\pi i} \int_{a-i\infty}^{a+i\infty} \frac{Q^*(s, r) \exp(sE^*) ds}{s}$$

where E^* is the energy at which the state count is to be calculated. The path of integration is a straight line parallel to the imaginary axis with abscissa a . The integrand is sharply peaked, and therefore can be approximated by the method of steepest descents. In view of eqn. (6), it is convenient to work with $\phi(z, r)$, the logarithm of the integrand, where

$z = \exp(-s)$:

$$\phi(z, r) = \ln Q^*(z, r) - \ln(\ln z^{-1}) - E^* \ln z$$

The integrand will have a saddle-point at some $z = \theta$, which is found by solving the transcendental equation $d\phi(z, r)/dz = 0$. If the integrand is expanded around θ to first order, it can be shown that

$$G^*(E^*, J, r) = \frac{Q^*(\theta, r)}{(\ln \theta^{-1}) \theta^{E^*} [2\pi \theta^2 \phi''(\theta, r)]^{1/2}}$$

where $\phi''(\theta, r)$ is the second derivative of $\phi(z, r)$ with respect to z evaluated at $z = \theta$. This expression provides an excellent approximation to $G^*(E^*, J, r)$ down to very low energies.

The reaction is assumed to take place under an effective potential $V_{\text{eff}}(r)$ which is approximated by treating the process $\text{CX}_3\text{O}_2 \rightarrow \text{CX}_3 + \text{O}_2$ as the fragmentation of a quasi-diatom AB into A + B. Thus

$$V_{\text{eff}}(r) = V(r) + E_{\text{rot}}(r_e/r)^2 \quad (7)$$

where $V(r)$ is the A-B potential and B_e in E_{rot} [eqn. (4)] is the equilibrium rotational constant of AB. In practice, the reactant CX_3O_2 is taken to be symmetric (or at least a near-symmetric) top, $V(r)$ is the $\text{X}_3\text{C}-\text{O}_2$ potential and B_e is the doubly-degenerate rotational constant of the symmetric top associated with the two-dimensional overall rotation with quantum number J . The quantum number K of the symmetric top (associated with rotation about the symmetry axis) is not conserved⁹ and the corresponding energy is therefore assumed to be distributable among the other (vibrational) degrees of freedom. Consequently the reactant partition function Q_m in eqn. (6) excludes the rotational degree of freedom corresponding to the two-dimensional J -rotor, but includes the one-dimensional degree of freedom of rotation of the K -rotor.

The bare interfragment potential $V(r)$ is approximated by the simple Morse function $V(r) = D[1 - \exp - \beta(r - r_e)]^2$ with β given by

$$\beta = 2\pi\nu(\mu/2D)^{1/2} \quad (8)$$

where ν is the frequency of the A-B bond undergoing rupture (*i.e.* the reaction coordinate), μ is the A-B reduced mass and D is the classical height of the bare potential barrier given by $D = \Delta H_0 + E_{\text{zm}} - E_{\text{zp}}$, with E_{zp} and E_{zm} the zero-point energies of the products and the molecule, respectively. The effective potential is corrected for zero-point energies at every r , which takes into account the progressive disappearance of the zero-point energy of the transitional modes as they are converted into rotations. The combined result of the interplay of the various elements of the calculation is that at specified E , and with $V_{\text{eff}}(r)$ of specified J , the state count $G^*(E^*, J, r)$ as a function of r is found to go through a minimum at some $r = r_{\min}$, which is where the numerator of eqn. (1) is evaluated.

In the present work, ν in eqn. (8) refers to the C-O vibrational mode throughout, but in contrast with the previous treatment⁸, μ will be taken as the reduced mass of the C and O atoms rather than the CX_3 and O_2 moieties. Consequently the reaction coordinate r refers to the C-O distance in $\text{CX}_3 \cdots \text{O}_2$.

The most important element of the variational approach is the switching function $S(r)$. Pauling's or Badger's rule¹⁰ predicts (and *ab initio* calculations confirm¹¹) that frequencies of vibrational modes decrease roughly exponentially at small and moderate bond extensions. If the frequencies of reactant transitional modes in a dissociation are made to decrease exponentially, the corresponding reactant state count $G(E, r)$

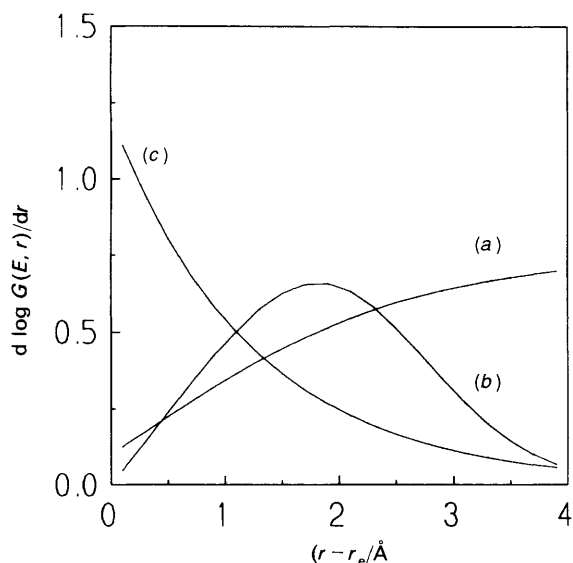


Fig. 1 Plot of $\partial \log G(E, r)/\partial r$ (in $\text{\AA}^{-1}$) vs. $r - r_e$ at $E = 1000 \text{ cm}^{-1}$ for the specific case $\text{CH}_4 \rightarrow \text{CH}_3 + \text{H}$ (a) $G(E, r)$ obtained by inversion of partition function for CH_4 in which transitional mode frequencies ν decline with distance r according to $\nu(r) = \nu_e \exp[-0.8691(r - r_e)]$. (b) $G(E, r)$ obtained by inversion of partition functions for CH_4 and CH_3 interpolated using $S(r) = 1 - \tanh[0.15(r - r_e)^2]$. (c) $G(E, r)$ obtained by inversion of partition functions for CH_4 and CH_3 interpolated using $S(r) = \exp[-0.75(r - r_e)]$. Subscript e refers to equilibrium properties of CH_4 .

at constant E has a slope $\partial G(E, r)/\partial r$ that varies with bond extension r as shown in Fig. 1(a). This curve shows physically reasonable behaviour at small r but not large r , when transitional mode frequencies drop exponentially to zero and the state count consequently approaches infinity.

If now we set up, for the same dissociation, a partition function interpolated according to eqn. (6), using an exponential switching function $S(r)$ similar to that used for frequencies in the case of Fig. 1(a), inversion yields a state count with slope $\partial G(E, r)/\partial r$ [Fig. 1(c)]. This curve is clearly of a completely different, and unphysical character at small r , although reasonable at large r since $\partial G(E, r)/\partial r$ must tend to zero with approach to the product state count which is finite.

What is needed, then, is a switching function $S(r)$ for the interpolation of partition functions *via* eqn. (6) that would produce upon inversion a state count that would behave as curve (a) at low r , and as curve (c) at high r . This is shown in Fig. 1(b) for the specific case $S(r) = 1 - a \tanh(r - r_e)^b$, but in fact any similarly S-shaped switching function would have the requisite qualitative behaviour.

The E in $G(E, r)$ represents randomizable energy, which for simplicity was assumed constant in the above arguments. In the strict context of eqn. (6) applied to an actual dissociation, E is the randomizable energy of the transition state (denoted as E^*), which is not constant but varies with r since the transition state 'sits' on the rising (attractive) part of the effective potential. The result is that under these conditions $G^*(E^*, J, r)$ as a function of r goes through a minimum. For a state count that behaves as in Fig. 1(c), this minimum is systematically found at larger r than in the case of the state count represented by the more physically reasonable curve (b). The practical consequence of such behaviour is to force all recombination rate constants to show systematically a positive temperature coefficient.

In the present instance we shall use, for simplicity, the one-parameter S-shaped (half-) Gaussian

$$S(r) = \exp[-c(r - r_e)^2]; \quad |r \geq r_e| \quad (9)$$

with its unique adjustable parameter c . While there are obviously other one-parameter S-shaped functions that could be used, the above Gaussian seems sufficient until such time as there should appear from other sources a more precise information about the nature of the switching function. Interestingly enough, a one-parameter Lorentzian suffers from the same defect as the simple exponential. The two-parameter hyperbolic tangent switching function mentioned above was also used previously.⁸ It produces results very similar to $S(r)$ of eqn. (9) and offers in principle a greater latitude of adjustment, but no greater insight.

Molecular Parameters

Exclusive of the reaction coordinate, in each of the reactions considered here there are 11 vibrational degrees of freedom of the reactant that correlate with seven vibrational and four rotational degrees of freedom of the products. There are thus 4 transitional vibrational modes in the reactant that correlate with free rotations of the products. Since the partition function of the transition state (Q^*) is obtained by interpolation of essentially the total partition functions of reactant and products, the individual transitional modes in the reactant do not have to be identified, only their number (or, equivalently, the number of rotational degrees of freedom in the products, less one [= the K -rotor]). Note, however, that the frequency of the reaction coordinate has to be known since it is used in calculating β in eqn. (8).

The K -rotor of the reactant correlates with the K -rotor of the CX_3 radical (*i.e.* the fifth rotational degree of freedom in the products), and the two-dimensional reactant J -rotor correlates with orbital movement of the products about their common centre of gravity. Angular momentum conservation is ignored. Since, as noted above, the J - and K -rotors of the reactant are treated differently, the corresponding rotational constants and symmetry numbers of the reactant have to be identified; no such distinction need be made in the products. In a true symmetric-top reactant this identification is no problem since the rotational constant of the J -rotor is doubly degenerate, and the rotational constant of the K -rotor is the remaining third rotational constant, either larger or smaller than the other two, depending on whether the top is prolate or oblate.

In our case the reactants are not quite symmetric tops, at least not in their equilibrium configuration, although it seems obvious that they will approach a true symmetric top as the C—O bond lengthens. This is illustrated in Fig. 2 for the case of CCl_3O_2 , using equilibrium atom coordinates as determined by MNDO, and allowing the C—O distance r to increase. The moment of inertia tensor is then resolved for every r to obtain the rotational constants A , B , C corresponding to the three principal axes of rotation. At $r = r_e$ the rotational constants are all different, but already from about $ca. 2 \text{ \AA}$ onwards the rotational constant B becomes indistinguishable from C . Considering that the transition state is generally located well above $2 \text{ \AA}$, this means that the stretched-out molecule becomes effectively a prolate symmetric top transition state with $A > B = C$. For simplicity, the strategy has been adopted to use for B_e in eqn. (4) the geometric mean of the equilibrium value of the two rotational constants that become equal with extension of the reaction coordinate. This was determined for each reaction by the above calculation, since it is not always obvious which two of the equilibrium rotational constants become equal with increasing interatomic distance r .

For the purpose of comparison with experiment, it was necessary to consider weak-collision effects. This was done approximately by writing for the effective collision frequency

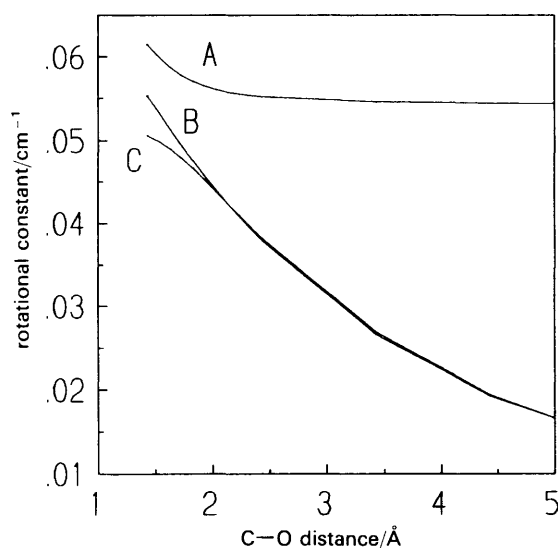


Fig. 2 Rotational constants of CCl_3O_2 as a function of C—O distance. The calculation neglects the deformation of the CCl_3 moiety and of the C—O—O angle with distance

ω in eqn. (2)

$$\omega = \beta_{wc} Z p \Omega \quad (10)$$

where β_{wc} is a weak-collision correction factor, assumed to be pressure- and energy-independent, Z is the hard-sphere colli-

sion number, p is pressure and Ω is the collision integral given by the approximate relation¹²

$$\Omega = [0.7 + 0.52 \log(kT/\epsilon)]^{-1} \quad (11)$$

where ϵ/k is the Lennard-Jones well depth in K.

Table 1 lists the pertinent molecular parameters for reactants and products of the recombination reactions considered, used as input for the calculations. The data for CH_3O_2 come from ref. 1, except for the rotational constants and collisional parameters which were taken from Patrick and Golden.¹³ Methyl radical data are from a standard compilation.¹⁴ Data for all other molecules and radicals were taken mostly from ref. 2; ref. 2(b) in particular gives the rationale behind the choice of parameter values, which was done in a systematic fashion. These are based on literature values, where available, and on MNDO calculations, mostly for the rotational constants. The CF_3 radical has been shown to be pyramidal,¹⁵ hence $\sigma = 3$. A recent study¹⁶ shows the CCl_3 radical also to be pyramidal, but subject to inversion doubling which introduces additional degeneracy, so that total $\sigma = 6$.

The best-characterized reactants and products are in the reactions $\text{CH}_3 + \text{O}_2 \rightarrow \text{CH}_3\text{O}_2$ and $\text{CCl}_3 + \text{O}_2 \rightarrow \text{CCl}_3\text{O}_2$, the parameter values of which are reasonably secure and for which there are available experimental values of the equilibrium constant K_{eq} over a range of temperatures.^{17,18} Such is not the case for the other two reactions considered; in particular, the ΔH_0 is only an estimate. Fortunately the dependence of the recombination rate constant on ΔH_0 is only

Table 1 Molecular parameters for reactants and products of the recombination reactions

molecule	reactant				
	CH_3O_2	CF_3O_2	CF_2ClO_2	CFCl_2O_2	CCl_3O_2
frequency/ cm^{-1}	2968 2968 2844 1440 1453 1414 1112 1183 1118 492 200 902	1292 1270 1129 728 596 584 1100 300 398 469 70 870	1280 1151 655 631 370 363 1100 268 310 503 72 965	1238 744 563 460 439 290 1100 215 283 497 78 998	815 761 523 370 326 272 1100 195 213 424 80 1017
reaction coordinate/ cm^{-1}	0.350	0.104	0.071	0.059	0.0506
rotational constants/ cm^{-1}	0.404 1.738	0.1045 0.182	0.089 0.119	0.073 0.080	0.0553 0.0615
symmetry number	1	1	1	1	1
$r_e/\text{\AA}$	1.46	1.4	1.43	1.43	1.43
$(\epsilon/k)/\text{K}$	340	188	253	333	322.7
$\sigma/\text{\AA}$	4	4.96	5.25	5.44	5.95
$\Delta H_0/\text{cm}^{-1}$	10733	12063	11201	10114	6855
products					
O_2 : frequency 1580 cm^{-1} , rotational constant 1.445 cm^{-1} , symmetry no. 2					
radical	CH_3	CF_3	CF_2Cl	CFCl_2	CCl_3
frequencies/ cm^{-1}	3162 3162 3044 1396 1396 607	1386 1384 971 660 530 525	1367 1221 637 557 447 366	1333 859 524 407 378 265	895 895 443 251 251 145
rotational constants/ cm^{-1}	4.79 9.58 9.58	0.181 0.3545 0.3545	0.109 0.153 0.368	0.074 0.107 0.235	0.054 0.108 0.108
symmetry numbers	6	3	2	2	6

indirect *via* D in eqn. (8), and therefore weak. The comparison of computed and experimental values of K_{eq} , where available, was used as a check on the molecular parameters of Table 1. Note that the equilibrium constant of all reactions involves an electronic degeneracy factor of 3.

The J -dependent recombination rate constant $k_{rec}(J)$ has to be calculated for a sufficiently large collection of J s to ensure that in every instance the J -averaged rate constant $\langle k_{rec} \rangle$ [from $\langle k_{diss} \rangle$ in eqn. (5)] is properly converged at all temperatures. For CH_3O_2 this required $0 \leq J \leq 184$, and $0 \leq J \leq 285$ for the other molecules.

Experimental data on recombination kinetics of the halogenated compounds were likewise taken from ref. 2. In the case of $\text{CFCl}_2 + \text{O}_2$ and $\text{CF}_3 + \text{O}_2$ these are updates of earlier publications;^{19,20} additional data in the latter case were taken from Ryan and Plumb.²¹ The $\text{CCl}_3 + \text{O}_2$ kinetic data are the subject of a recent detailed account.^{2(b)} The $\text{CH}_3 + \text{O}_2$ experimental data come from several sources: the 298 K data at lower pressures are those of Pilling and Smith,²² and at higher pressures those of Cobos *et al.*;²³ at all higher temperatures the data are from ref. 1. Experimental data for the title reactions are summarized in Table 4 (see later).

Results

In addition to molecular data of Table 1, the calculation requires as input the value of c , the adjustable parameter of the switching function of eqn. (9). For a given reaction, this was obtained by choosing a unique c that gave a best fit of $\langle k_{rec} \rangle$ to experiment for the entire range of pressures and temperatures covered. Where experimental data were mostly in the fall-off range, best fit meant essentially a fit to the lowest- and highest-pressure experimental point at each temperature, using the weak-collision factor β_{wc} to shift the calculated fall-off curve along the pressure axis for best coincidence with all the available experimental points.

A graphical representation of the experimental data (symbols) and calculated data (continuous lines) on the fall-off of the recombination rate constants for the title reactions is shown in Fig. 3 and 4. Table 2 gives values of the weak-collision parameter β_{wc} determined in the above fashion, as

Table 2 Summary of weak-collision data

T/K	β_{wc}	$\langle \Delta E \rangle_{\text{total}}/\text{cm}^{-1}$	$\langle \Delta E \rangle_{\text{down}}/\text{cm}^{-1}$
$\text{CH}_3 + \text{O}_2 + \text{M}$			
298	0.32	173	306
334	0.27	151	290
474	0.13	82	229
582	0.09	67	225
$\text{CF}_3 + \text{O}_2 + \text{M}$			
233	0.35	156	263
298	0.2	87	194
373	0.12	58	166
295 ^a	0.12	44	126
$\text{CFCl}_2 + \text{O}_2 + \text{M}$			
233	0.22	77	165
298	0.12	46	132
373	0.08	37	129
$\text{CCl}_3 + \text{O}_2 + \text{M}$			
233	0.4	215	340
298	0.23	119	247
373	0.15	87	229

All M refer to N_2 except for $\text{CH}_3 + \text{O}_2$ where $\text{M} = \text{Ar}$. All $\langle \Delta E \rangle$ are negative. ^a Refers to $\text{M} = \text{He}$.

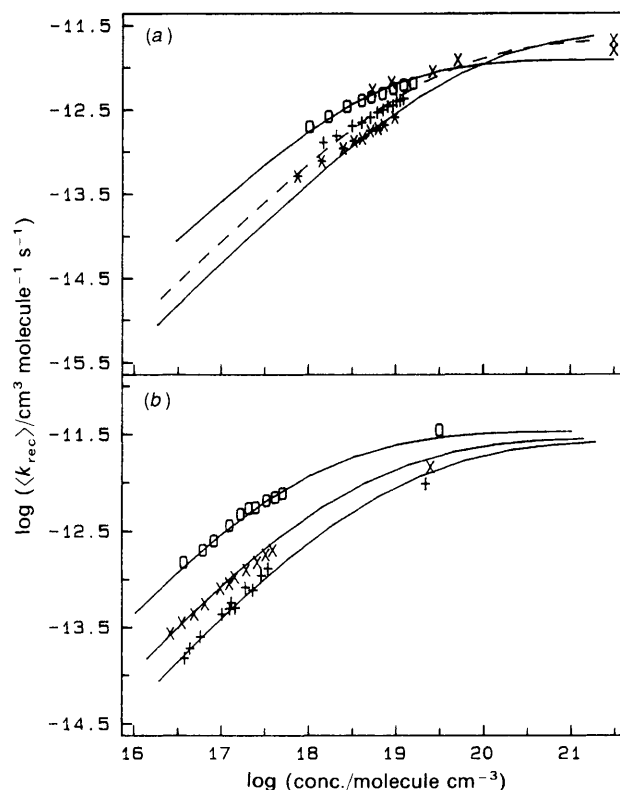


Fig. 3 Comparison of experimental and calculated recombination rate constant fall-off at several temperatures. (a) $\text{CH}_3 + \text{O}_2$. (○), 298 K data of ref. 22; ×, 298 K data of ref. 23. All other data are those of ref. 1. +, 474 K; *, 582 K. Argon heat bath gas throughout. (b) $\text{CCl}_3 + \text{O}_2$. ○, 233 K; ×, 298 K; +, 333 K. Experimental data of ref. 2(b) for nitrogen heat bath gas

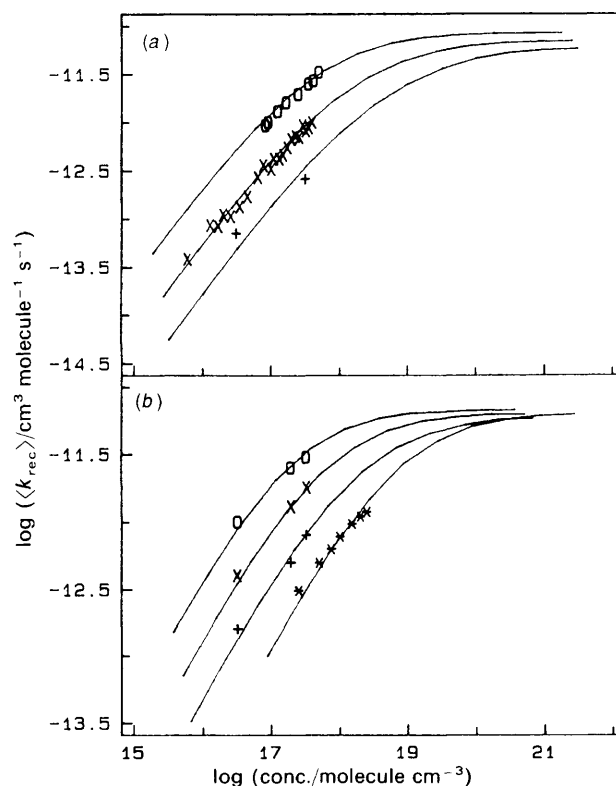


Fig. 4 Comparison of experimental and calculated recombination rate constant fall-off at several temperatures. (a) $\text{CFCl}_2 + \text{O}_2$. ○, 233 K; ×, 298 K; +, 373 K. Experimental data of ref. 2(a) for nitrogen heat bath gas. (b) $\text{CF}_3 + \text{O}_2$. ○, 233 K; ×, 298 K; +, 373 K. Experimental data of ref. 2(a) for nitrogen heat bath gas. *, 295 K: experimental data of Ryan and Plumb²¹ with helium heat bath gas, displaced for clarity by one log unit to higher concentrations

Table 3 Summary of calculated rate constants $\langle k_{\text{rec}} \rangle = a(T/298)^b$

reactants	<i>c</i>	$\langle k_{\text{rec}, \infty} \rangle / \text{cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$		$\langle k_{\text{rec}, 0} \rangle / \text{cm}^6 \text{ molecule}^{-2} \text{ s}^{-1}$	
		<i>a</i>	<i>b</i>	<i>a</i>	<i>b</i>
CH ₃ + O ₂	0.195	1.23×10^{-12}	+1.19	3.02×10^{-31}	-2.68
CF ₃ + O ₂	0.23	6.34×10^{-12}	-0.27	1.55×10^{-29}	-4.78
CF ₂ Cl + O ₂ ^b	0.296	7.11×10^{-12}	-0.56	1.44×10^{-29}	-5.19
CFCl ₂ + O ₂	0.40	7.01×10^{-12}	-0.77	6.16×10^{-30}	-5.61
CCl ₃ + O ₂	0.535	2.95×10^{-12}	-0.63	1.78×10^{-30}	-6.41

c is the parameter in the switching function $S(r) = \exp[-c(r - r_e)^2]$. The temperature range considered is that shown in Table 2. ^a $\langle k_{\text{rec}, 0} \rangle$ is the limiting low-pressure weak-collision rate constant with β_{wc} factored in. Collision partner is M = N₂ throughout except for CH₃O₂ where M = Ar. ^b predicted (see Discussion).

well as average collisional energy transfers calculated from the approximate relations²⁴

$$\langle \Delta E \rangle_{\text{down}} = \frac{\beta_{\text{wc}}^{1/2} k T F_E}{1 - \beta_{\text{wc}}^{1/2}}; \quad \langle \Delta E \rangle_{\text{total}} = -\langle \Delta E \rangle_{\text{down}} \beta_{\text{wc}}^{1/2} \quad (12)$$

Table 3 lists switching function parameter *c* used to obtain the calculated data in Fig. 3 and 4, together with the least-squares fit of the limiting high- and low-pressure recombination rate constants according to the simple power law $\langle k_{\text{rec}} \rangle = a(T/298)^b$. The actual temperature dependence of $\langle k_{\text{rec}} \rangle$ is somewhat more complicated than this, but over a limited temperature range involved the power law is a reasonably good representation. The listed values of $\langle k_{\text{rec}, 0} \rangle$ are based on the weak-collision factor β_{wc} from Table 2. Note that these limiting rate constants are not extrapolations but results calculated from eqn. (2) for $\omega \rightarrow \infty$ and $\omega \rightarrow 0$, respectively.

Discussion

It is useful to start with the limiting rate constants in Tables 3 and 4 for the best-characterized reactions. For CH₃ + O₂, the experimental $k_{\text{exp}, \infty}$ of ref. 1 is virtually identical to $\langle k_{\text{rec}, \infty} \rangle$ in Table 3 which is calculated from data between 298 and 582 K according to the simple power law in *T*. Over such a wide temperature range a more accurate representation of the calculations is $\langle k_{\text{rec}, \infty} \rangle = 8.94 \times 10^{-16} T^{1.23} \exp(-74.64/T)$. At the low-pressure limit, $k_{\text{exp}, 0}$ of ref. 1 is somewhat larger with a larger negative temperature coefficient than the calculated $\langle k_{\text{rec}, 0} \rangle$. These results may be compared with SACM calculations of Cobos *et al.*²³ who report $k_{\text{rec}, \infty} = 2.2 \times 10^{-12} (T/300)^{0.94}$ and $k_{\text{rec}, 0} = (7 \pm 1) \times 10^{-31} (T/300)^{-3.0}$ in the range 200–400 K, the latter rate constant assuming a temperature-independent $\langle \Delta E \rangle_{\text{total}}$.

For CCl₃ + O₂, both experimental $k_{\text{exp}, \infty}$ and $k_{\text{exp}, 0}$ are quite close to the corresponding calculated $\langle k_{\text{rec}, \infty} \rangle$ and $\langle k_{\text{rec}, 0} \rangle$ in Table 3. While the 'experimental' values of $k_{\text{exp}, \infty}$ and $k_{\text{exp}, 0}$ for these two reactions represent in fact extrapolations by two different techniques, and therefore are subject to caution, they show unequivocally that the temperature coefficient of $k_{\text{exp}, \infty}$ is positive in CH₃ + O₂, and negative in CCl₃ + O₂, a fact well reproduced by the present calculation.

Turning now to the actual fall-off data for these two reactions shown in Fig. 3, observe that in the case of CH₃ + O₂ at 298 K, the calculated high-pressure limit falls below the highest-pressure (148 bar) experimental result of ref. 23, but at lower pressures, and at all temperatures, the fall-off is reasonably well rendered (fall-off at 334 K is not shown to avoid crowding in Fig. 2). Unfortunately there are no high-pressure experimental results available above 298 K. In the case of CCl₃ + O₂, there is at least one higher-pressure datum (760 Torr) at each temperature, which allows a fairly good characterization of the fall-off which the calculation reproduces reasonably well. The weak-collision factors β_{wc} used in these calculations and the corresponding energy transfer data listed in Table 2 show a downward trend with temperature.

In the case of the limiting rate constants for the other two reactions, CF₃ + O₂ and CFCl₂ + O₂, 'experiment' (*i.e.* published² extrapolation of experimental fall-off data shown in Table 4) is less precise and gives the temperature coefficient of $k_{\text{exp}, \infty}$ only as $\propto T^{0 \pm 1}$ due to lack of high-pressure data in both cases, with smaller uncertainty in $k_{\text{exp}, 0}$.

It is worthwhile at this point examining more closely what happens when fitting limited experimental data by MVIPF. The general effect of the adjustable constant *c* is the following: increasing *c* in a given reaction produces a microcanonical rate constant for dissociation that increases with energy more steeply, which translates into a larger negative coefficient of the thermal rate constant for recombination, and moves the recombination fall-off to lower pressures, increasing its curvature and increasing $\langle k_{\text{rec}, \infty} \rangle$. In the absence of high-pressure experimental data, the fit therefore concerns the absolute position of the fall-off plot on the pressure axis and its curvature. If *c* is too small, calculated fall-off will be positioned at such high pressures that to make theory and experiment coincide would require unreasonably large values of β_{wc} , with a slope of the fall-off plot not sufficiently steep. If *c* is too large, calculated fall-off will occur at such low pressures as to require unreasonably small values of β_{wc} to bring it into coincidence with experiment, with a slope that is too steep. The obvious repercussions for $\langle k_{\text{rec}, \infty} \rangle$ are that in the former case $\langle k_{\text{rec}, \infty} \rangle$ and its temperature coefficient are too small, and in the latter case both are too large. This is illustrated in Table 5 using the recombination CF₃ + O₂ as the worst-case scenario. For *c* = 0.15, the calculated $\langle k_{\text{rec}, \infty} \rangle$

Table 4 Summary of experimental rate constants $k_{\text{exp}} = a(T/298)^b$

reactants	ref.	$k_{\text{exp}, \infty} / \text{cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$		$k_{\text{exp}, 0} / \text{cm}^6 \text{ molecule}^{-2} \text{ s}^{-1}$	
		<i>a</i>	<i>b</i>	<i>a</i>	<i>b</i>
CH ₃ + O ₂ ^a	1	$1.2 \pm 0.2 \times 10^{-12}$	+1.2 ± 0.04	$1 \pm 0.3 \times 10^{-30}$	-3.3 ± 0.4
CF ₃ + O ₂	2(a) }	$\approx 9 \pm 3 \times 10^{-12}$	0 ± 1	$1.9 \pm 0.2 \times 10^{-29}$	-4.7 ± 0.5
CFCl ₂ + O ₂	2(a) }			$5.5 \pm 1 \times 10^{-30}$	-6 ± 0.5
CCl ₃ + O ₂	2(b)	$3.2 \pm 0.7 \times 10^{-12}$	-1.2 ± 0.4	$1.6 \pm 0.3 \times 10^{-30}$	-6.3 ± 0.5

^a These results conform to $k_{\text{exp}} = a(T/300)^b$

Table 5 Effect of adjustable parameter c on $\langle k_{\text{rec}, \infty} \rangle = a(T/298)^b$ $\text{cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ and on β_{wc} in the recombination $\text{CF}_3 + \text{O}_2$

c	a	b	$\beta_{\text{wc}}(233 \text{ K})$
0.15	2.07×10^{-12}	+0.15	(1.7)
0.20	4.95×10^{-12}	−0.18	0.5
0.25	1.03×10^{-11}	−0.42	0.25
0.30	1.93×10^{-11}	−0.61	0.15

at 233 K is so low that it is lower than the experimental result at $3.2 \times 10^{17} \text{ molecule cm}^{-3}$ ($= 7.7 \text{ Torr}$), requiring, in addition, an impossible β_{wc} . For $c = 0.30$, the calculated fall-off curve is so steep that it cannot be fitted to more than one experimental point at any one temperature. This leaves the best fit somewhere between $c = 0.20$ and $c = 0.25$, with $c = 0.23$ of Table 3 and Fig. 4 as the best compromise. From Table 5 a 0.05 difference in c produces roughly a difference of factor of 2 in $\langle k_{\text{rec}, \infty} \rangle$. This means that in this the worst case the calculated $\langle k_{\text{rec}, \infty} \rangle$ quoted in Table 3 should be good to within a factor of two, and certainly better than within a factor of four, if one assumes pessimistically, best compromise notwithstanding, that $0.20 \leq c \leq 0.30$. Note that the calculation leaves absolutely no doubt that the temperature coefficient of $\langle k_{\text{rec}, \infty} \rangle$ is negative.

Concerning the fall-off data (Fig. 3 and 4) and the energy-transfer parameters that can be deduced from the calculations (Table 2), one sees a virtually uniform $\text{ca. } T^{-2}$ temperature dependence of β_{wc} in all four reactions, so that both $\langle \Delta E \rangle_{\text{down}}$ and $\langle \Delta E \rangle_{\text{total}}$ likewise decrease with temperature in absolute value, but less strongly. All the $\langle \Delta E \rangle_{\text{down}}$ are in the 200–300 cm^{-1} range, which seems reasonable for $\text{M} = \text{Ar}$ or N_2 ; note that $\langle \Delta E \rangle_{\text{down}}$ for $\text{M} = \text{He}$ at 295 K in the case of $\text{CF}_3 + \text{O}_2$ is significantly lower, as might be expected. By solving the weak-collision master equation, Pilling *et al.*¹ deduced $\langle \Delta E \rangle_{\text{down}} \approx -285 \text{ cm}^{-1}$ for $\text{CH}_3 + \text{O}_2$, independent of temperature, not too far from the value calculated in Table 2, which, however, is seen to decrease slightly with temperature. On the other hand, Cobos *et al.*²³ report for this reaction $\langle \Delta E \rangle_{\text{total}} = -42 \text{ cm}^{-1}$ at 298 K (which amounts to $\langle \Delta E \rangle_{\text{down}} = -125 \text{ cm}^{-1}$).

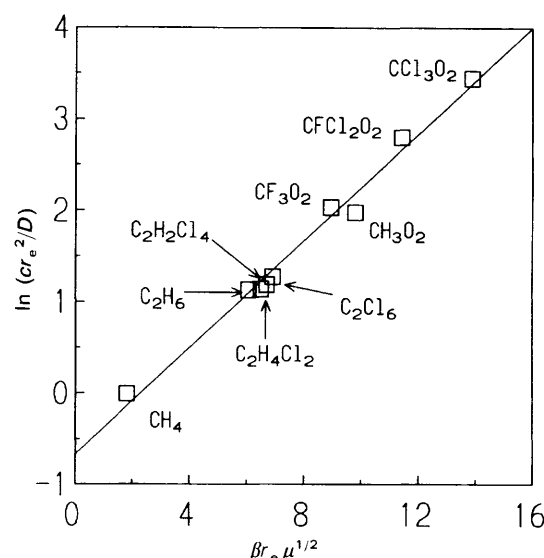
In contrast with reactions where experimental data are mostly below the high-pressure limit, radical self-combinations listed in Table 6 and discussed more fully in the accompanying paper,²⁵ are essentially at the high-pressure limit since there is no experimentally detectable fall-off at temperatures below *ca.* 350 K. The one exception is $\text{CH}_3 + \text{CH}_3$, but here the fall-off has been subjected to careful analysis²⁶ of extensive experimental data²⁷ in order to provide a secure value for $k_{\text{rec}, \infty}$. Hence the experimental $k_{\text{exp}, \infty}$ of all the recombinations in Table 6 provide unambiguous reference for fitting the adjustable constant c .

It remains to be demonstrated that the parameter c , as adjusted for the various reactions, actually conforms to a pre-

Table 6 Summary of calculated rate constants $\langle k_{\text{rec}, \infty} \rangle = a(T/298)^b$ for high-pressure self-combination of radicals^a

radical	c	$\langle k_{\text{rec}, \infty} \rangle \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$	
		a	b
CH_3	0.204	6.18×10^{-11}	−0.34
CH_2Cl	0.200	2.91×10^{-11}	−0.72
CHCl_2	0.195	1.01×10^{-11}	−1.31
CCl_3	0.1725	3.74×10^{-12}	−1.48

^a Data from Table 6 in ref. 25. c is the parameter in the switching function $S(r) = \exp[-c(r - r_e)^2]$. The temperature range considered is 253–363 K.

**Fig. 5** Semi-logarithmic plot of adjustable parameter c in switching function of eqn. (9) for all radical recombination reactions studied by MVIPF. The straight line represents eqn. (13). Each recombination is identified by the stable product molecule

dictable pattern. Fig. 5 shows a semi-logarithmic plot of the Gaussian parameter c for all the reactions studied so far by MVIPF, *i.e.* the four reactions described in this work, plus the radical recombinations $\text{CH}_2\text{Cl} + \text{CH}_2\text{Cl}$, $\text{CHCl}_2 + \text{CHCl}_2$ and $\text{CCl}_3 + \text{CCl}_3$ of Table 6, as well as the recombinations $\text{CH}_3 + \text{H}$ and $\text{CH}_3 + \text{CH}_3$ of ref. 8, where the Gaussian parameter c was recalculated to conform to the present definition of μ in eqn. (8). The plot is of the variables $\ln cr_e^2/D$ vs. $\beta r_e \mu^{1/2}$, where β , D , and μ are the Morse parameters in eqn. (8), the last two in atomic units (for energy the unit is hartree ($1 E_h = 2.19487 \text{ cm}^{-1}$)). It can be seen that virtually all the c s fall on the same straight line, the equation of which is

$$\ln cr_e^2/D = (-0.6713 \pm 0.069) + (0.2920 \pm 0.013)\beta r_e \mu^{1/2} \quad (13)$$

This is an interesting result that connects the values of the parameter c for reactions as disparate in character as $\text{CH}_3 + \text{H}$ and $\text{CCl}_3 + \text{O}_2$, or as different as $\text{CH}_3 + \text{O}_2$ and $\text{CH}_3 + \text{CH}_3$, the first with a positive temperature coefficient, and the second with a negative temperature coefficient.

In a series of similar reactions, eqn. (13) now allows one in principle to fix the approximate value of c without reference to actual experimental data, on the basis of molecular parameters of reactants and product alone. To assess the significance of this result, it is worth recalling that eqn. (13) represents a fit of c to experimental data of a total of nine reactions, of which at most two ($\text{CFCl}_2 + \text{O}_2$ and $\text{CF}_3 + \text{O}_2$) may be considered as providing experimental data limited in scope and therefore insufficient for a precise determination of constant c .

Note that while $\langle k_{\text{rec}} \rangle$ depends only weakly on ΔH_0 , c in eqn. (13) depends on ΔH_0 directly *via* D ; in other words, the prediction of parameter c *via* eqn. (13) depends on how well the thermochemistry of the reaction is known. Since c depends also on other parameters of the potential in eqn. (8), there is thus clearly a connection between c and the potential. Pending further work, it is only prudent, for the time being, to treat eqn. (13) as a purely empirical relation, the physical significance of which remains to be determined.

The indicated variance in eqn. (13) amounts to approximately a 10% error in c , which, in practical terms, amounts roughly to an error of a factor of two in $\langle k_{\text{rec}} \rangle$. Hence it

should be possible, using MVIPIF in a series of similar reactions, to predict the rate constant of a recombination reaction not yet measured within a factor of two. This demonstrates the practical utility of MVIPIF, an adjustable parameter notwithstanding, a parameter that on the basis of the above relation ceases to be merely an arbitrary *ad hoc* factor.

As an example, take the recombination $\text{CF}_2\text{Cl} + \text{O}_2$ for which no experimental data are available at present for instrumental reasons. Using molecular data listed in Table 1 and parameter c obtained from eqn. (13), Table 3 gives calculated results for the high- and low-pressure limiting recombination rate constants. The rate constant $\langle k_{\text{rec},0} \rangle$ includes a weak-collision factor β_{wc} obtained by interpolation between those of $\text{CF}_3 + \text{O}_2$ and $\text{CFCl}_2 + \text{O}_2$. Future measurements will tell how well or how poorly these calculations compare with experiment.

The results of the calculations in Tables 2 and 3 suggest that the $\text{CH}_3 + \text{O}_2$ reaction, contrary to some expressed views,^{1,23} is not much different from the other recombinations of the $\text{CX}_3 + \text{O}_2$ type, both by reason of the magnitude of k_∞ , which is comparable to k_∞ of the other reactions of the same type, and by reason of its temperature coefficient, since the same MVIPIF formalism is able to reproduce a positive temperature coefficient in this particular case, and a negative one in all the others, without the need to invoke a special feature, *e.g.* a small barrier in the exit channel.

Similarly the fact that in Fig. 5 the value of c corresponding to $\text{CH}_3 + \text{O}_2$ seems to be somewhat out of line with the other values does not necessarily invalidate the above conclusion that this reaction is basically no different from all the others. In fact, if the MVIPIF calculation were done with a value of c that lies exactly on the straight line in Fig. 5, $\text{CH}_3 + \text{O}_2$ would still show a positive temperature coefficient.

If $\text{CH}_3 + \text{O}_2$ is a reaction no different from all the others, then the obvious intriguing question is what causes it to have nevertheless a positive temperature coefficient, unlike the other $\text{CX}_3 + \text{O}_2$ -type reactions, in particular unlike $\text{CF}_3 + \text{O}_2$ which is in many ways similar. This is best understood by examining first the rotationless canonical recombination rate constant $k_{\text{rec},\infty}(J=0)$ which can be written

$$k_{\text{rec},\infty}(J=0) \propto \frac{kT}{h} \frac{Q_{\text{M}}^S Q_{\text{P}}^{1-S}}{Q_{\text{M}}} \exp(-E_0/kT) \times \frac{Q_{\text{M}} kT/B_{\text{c}}}{T^{3/2} Q_{\text{P}}} \exp(\Delta H_0/kT) \quad (14)$$

where both Q_{M} and Q_{m} are the vibrational + K -rotor partition functions for the reactant, the latter excluding the reaction coordinate frequency, Q_{P} is the total (vibrational + rotational) partition function for the products, kT/B_{c} is the reactant J -rotor partition function, S is the value of the switching function of eqn. (9) evaluated at $r = r_{\text{min}}$, and E_0 is the critical energy for decomposition at r_{min} . The first factor on the r.h.s. of eqn. (14) is the rate constant for decomposition, and the second factor is the equilibrium constant for decomposition, where the factor $T^{3/2}$ in the denominator arises from the translational partition function. Eqn. (14) simplifies to

$$k_{\text{rec},\infty}(J=0) \propto \left\{ T^{1/2} \left(\frac{Q_{\text{m}}}{Q_{\text{P}}} \right)^S \right\} \left\{ \exp[-(E_0 - \Delta H_0)/kT] \right\} \quad (15)$$

A B

The temperature dependence of the factors A and B is shown in Table 7 separately for $\text{CH}_3 + \text{O}_2$ and $\text{CF}_3 + \text{O}_2$. Rotational averaging, *i.e.* passing from $k_{\text{rec},\infty}(J=0)$ to $\langle k_{\text{rec},\infty} \rangle$ introduces an additional *ca.* $T^{-0.22}$ dependence, for a total of *ca.* $T^{+1.24}$ and *ca.* $T^{-0.25}$ for the two reactions,

Table 7 Temperature coefficients of factors A and B in eqn. (15)

reaction	$\text{CH}_3 + \text{O}_2$	$\text{CF}_3 + \text{O}_2$
A	$T^{-0.56}$	$T^{-0.39}$
B	$T^{+1.98}$	$T^{+0.36}$
total	$T^{+1.42}$	$T^{-0.03}$

Temperature coefficients determined in the range 300–600 K.

respectively. It can be seen that in these two cases the determining factor for the sign of the temperature coefficient is B , *i.e.* the difference $E_0 - \Delta H_0$.

The enthalpy ΔH_0 (a primary input assumed to be available from other information) contains the energy change due to electronic rearrangement between reactant and products, whereas E_0 is the similar energy change between reactant and transition state. In a given reaction, both ΔH_0 and E_0 refer to the same zero-point energy-corrected potential, but at different locations: ΔH_0 at $r \rightarrow \infty$, and E_0 at $r = r_{\text{min}}$. In the two reactions considered, r_{min} is very similar (*ca.* 3 Å at *ca.* 300 K). Thus the different temperature coefficient of B in the two reactions can be ascribed principally to the difference between zero-point energies of reactant and products, which is large for $\text{CH}_3 + \text{O}_2$ ($E_{\text{zm}} - E_{\text{zp}} = 1873 \text{ cm}^{-1}$), and small for $\text{CF}_3 + \text{O}_2$ ($E_{\text{zm}} - E_{\text{zp}} = 885 \text{ cm}^{-1}$), the result of relatively high vibrational frequencies in $\text{CH}_3 + \text{O}_2$, and relatively low ones in $\text{CF}_3 + \text{O}_2$.

As a rough guide, it can be said that, everything else being equal (*i.e.* in a series of similar reactions), the reactant with the lower vibrational frequencies (*i.e.* one with heavy substituents) will tend to have a more negative temperature coefficient of the rate constant for recombination. This is particularly well borne out by the calculations on the recombinations $\text{CX}_3 + \text{CX}_3 \rightarrow \text{C}_2\text{X}_6$ in Table 6.

Summary

Given the nature and aims of MVIPIF, there are obviously many gross simplifications. The strategem is to identify, and then to simplify what is not essential, and at the same time treat as correctly as possible what is important. By the evidence, the important element appears to be that the system initial (reactant) state count evolve to the correct final (product) count, with careful attention to the evolution of zero-point energies, notably the disappearance of zero-point energies of vibrators in the process of transformation into free rotors. This concerns essentially the degrees of freedom orthogonal to the reaction coordinate, and at first sight it seems preposterous to suggest that any information concerning this evolution is contained in the reaction coordinate itself, yet this is precisely what Fig. 5 and eqn. (13) suggest: the adjustable parameter c which governs the reactant $\rightarrow$ products evolution appears to depend entirely on properties of the (admittedly crude) potential along the reaction coordinate.

To be sure, this dependence, as expressed by eqn. (13), holds for the particular combination of switching function and potential used in MVIPIF. There are obviously many other potentials that could have been used, but given the constraint of limited data, the choice is limited. Similarly, other forms of the switching function come to mind, but it is shown earlier that it has to have the general form of an S-shaped function. Given this constraint, the potential plays only a subsidiary role, in the sense that minor changes or adjustments in potential parameters require a change in the numerical value of the adjustable parameter c in order to fit experiment, without, however, significantly affecting the final

result, *i.e.* the numerical value of $\langle k_{\text{rec}} \rangle$ and, in particular, its temperature coefficient.

This is rather fortunate since even the application of the simple Morse potential is not entirely straightforward: for example, there is an uncertainty in what to use for the reduced mass μ in eqn. (8). In the present instance the choice was made to use the C—O reduced mass, prompted by the observation that the C—C bond frequency in CX_3CX_3 compounds is virtually independent of the masses of the CX_3 moieties in self-association (ref. 25, Table 4), or that the C—O bond frequency is nearly independent of the masses of CX_3 and O_2 moieties in $\text{CX}_3 + \text{O}_2$ associations (cf. Table 1). If μ is defined in terms of moieties rather than atoms, the numerical value of the adjustable parameter c has to be increased roughly in proportion of the corresponding reduced masses, but without significant effect on $\langle k_{\text{rec}} \rangle$.

Another simplification is involved in the fabrication of the effective potential in eqn. (7). The rotational part amounts to assuming, in effect, that the J -rotor rotational constant declines with distance according to $B_e(r_e/r)^2$, which is a stronger decline than that suggested by Fig. 2. Now if E_{rot} in eqn. (7) were defined in terms of the presumably more reasonable r -dependent rotational constant shown in Fig. 1 (ignoring the fact that equilibrium atom coordinates are not generally available for most reactants), a fit to experiment would require a change in the adjustable parameter c , as above, but again without changing substantially the final result $\langle k_{\text{rec}} \rangle$.

The point is that if all reactions are treated in the same approximate, but systematic fashion, there emerges an internal coherence in the values of the adjustable parameter c for the various reactions, as represented by Fig. 5 and eqn. (13).

The treatment of weak collisions by means of the weak-collision correction factor β_{wc} is obviously a considerable short-cut, made in the interest of simplification, as is the translation of β_{wc} into energy transfer parameters $\langle \Delta E \rangle$ via eqn. (12). Therefore the trends in Table 2 are likely to be more significant than the actual numerical values.

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